



# MBOA Market - Technical Diagrams

Complete set of UML diagrams, flowcharts, and architecture diagrams for the defense memoir

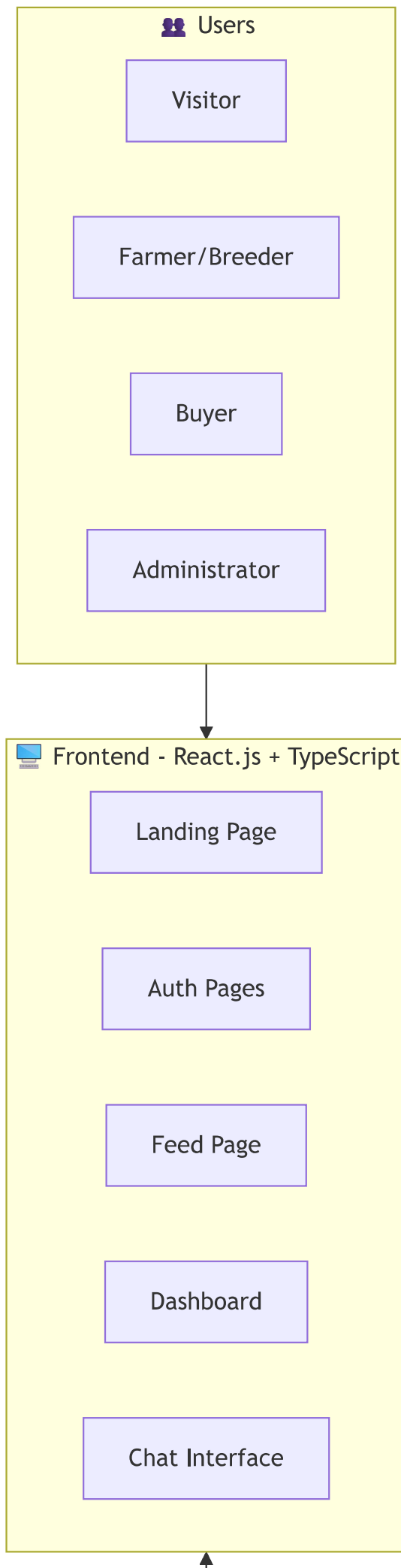
# 1 System Architecture Diagram

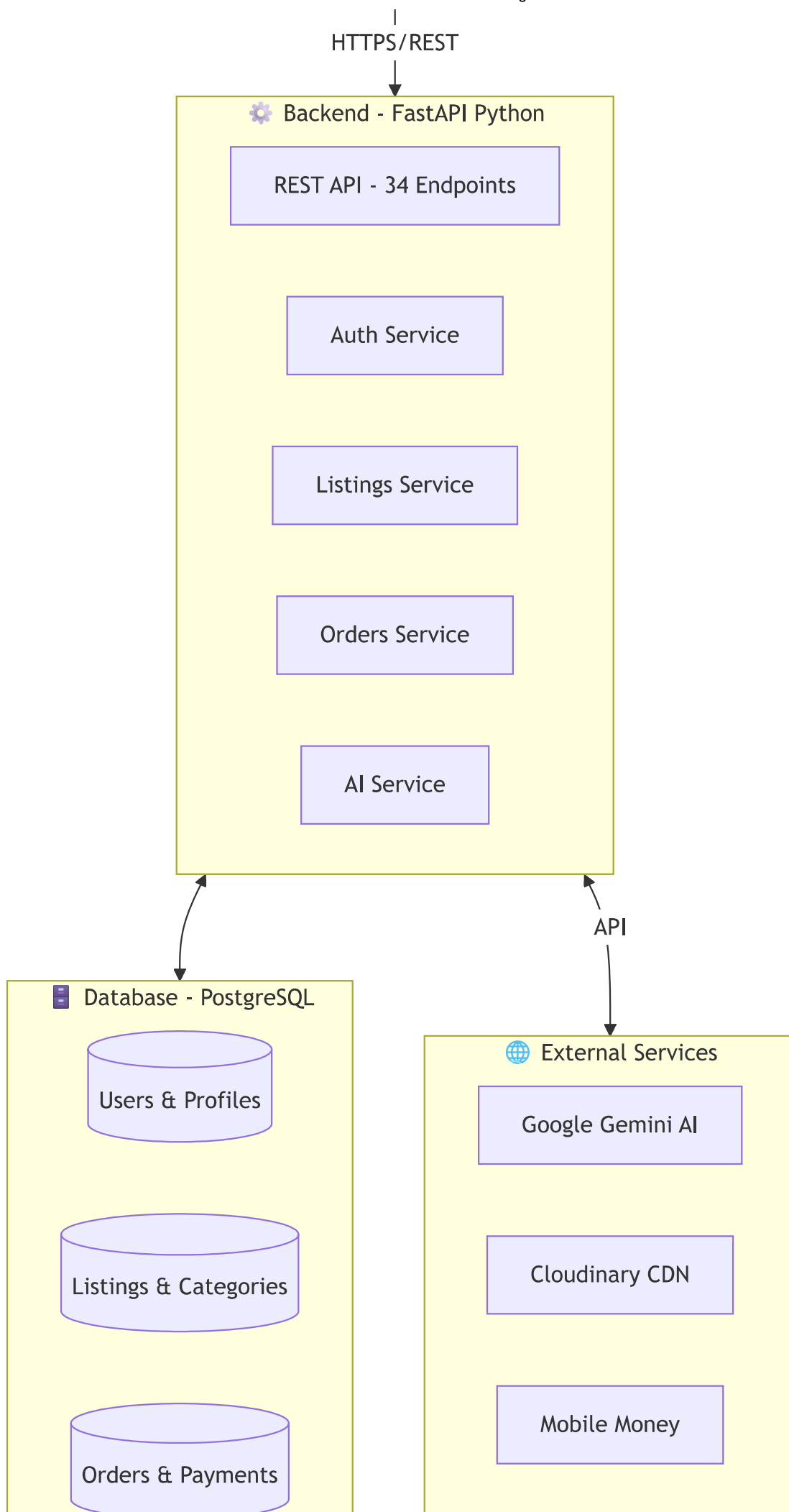
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## Introduction

The MBOA Market platform follows a modern three-tier architecture that separates concerns between presentation, business logic, and data persistence. This architectural choice ensures scalability, maintainability, and security. The system integrates with external services to provide AI-powered assistance, media storage, and mobile payment capabilities.









### ***Figure 3.1: MBOA Market System Architecture Overview***

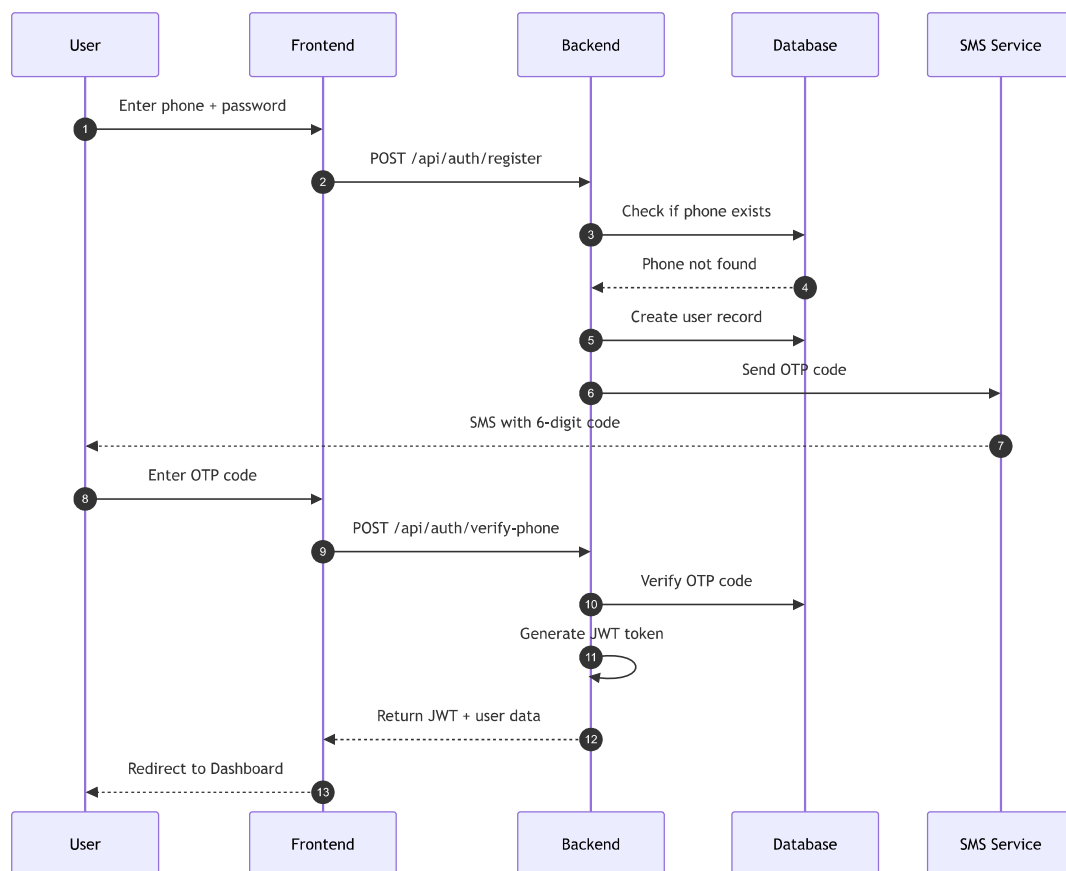
#### **Explanation**

The architecture consists of four main layers: (1) User Layer with four actor types, (2) Frontend Layer built with React.js and TypeScript, (3) Backend Layer using FastAPI with 34 RESTful endpoints, and (4) Database Layer with PostgreSQL storing 33 tables. External services include Google Gemini for AI, Cloudinary for media, and Mobile Money for payments.

## 2 Sequence Diagram - Authentication

### Introduction

The authentication process is a critical component of the MBOA Market platform, ensuring secure access while maintaining a user-friendly experience. The system implements phone-based authentication with OTP verification, suited for the Cameroonian market where phone numbers are more commonly used than email addresses.



**Figure 3.2: Sequence Diagram - User Authentication Process**

### Explanation

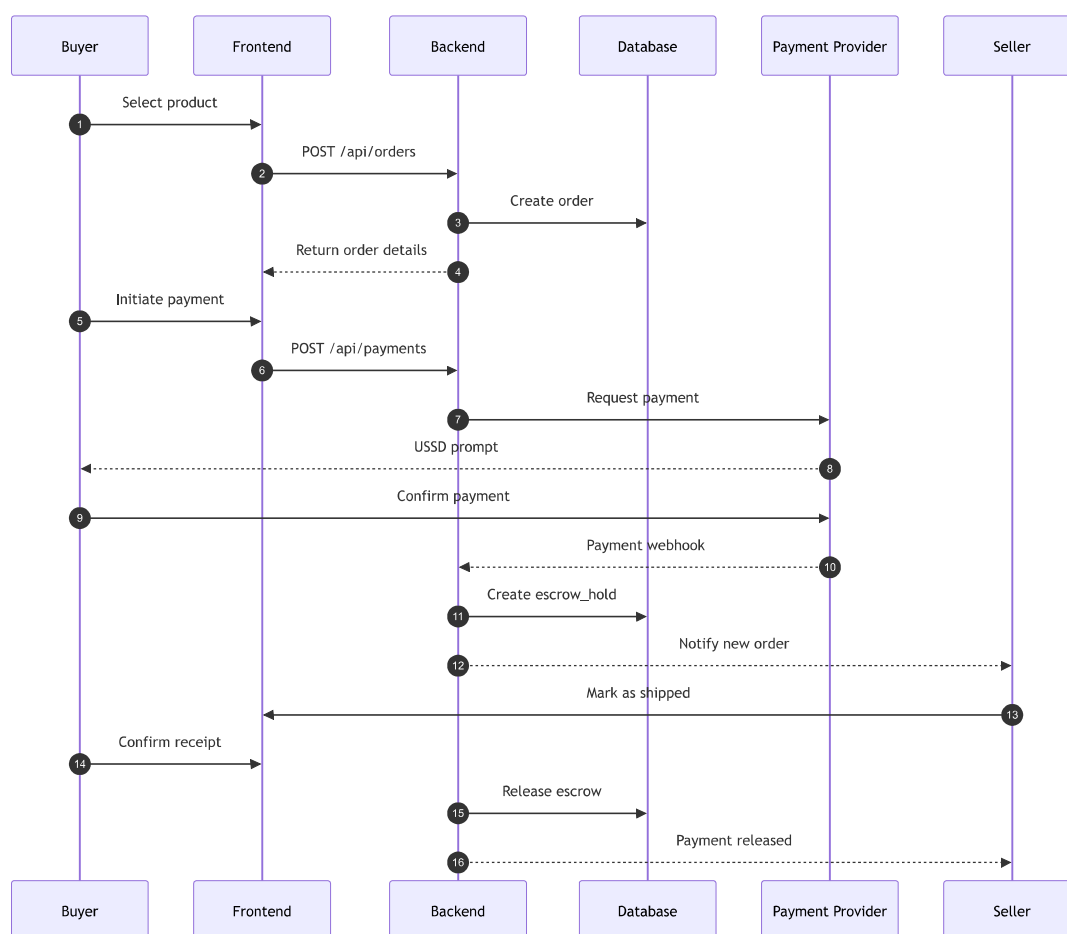
The authentication sequence involves: (1) User enters credentials, (2) Backend validates and creates user record, (3) OTP is sent via SMS, (4) User verifies OTP, (5) JWT token is generated for session management. This approach provides security

through password hashing, auditability through login logging, and a familiar phone-based experience for local users.

### 3 Sequence Diagram - Order Process

#### Introduction

The order process in MBOA Market follows a secure escrow-based model that protects both buyers and sellers. This diagram illustrates the complete lifecycle of an order, from product selection to payment confirmation and delivery fulfillment.



**Figure 3.3: Sequence Diagram - Order and Payment Process**

#### Explanation

The order process consists of three phases: (1) Order Creation with fee calculation, (2) Payment Processing via Mobile Money with escrow protection, and (3)

Fulfillment with delivery confirmation. Key security features include escrow protection, idempotency for payments, audit trails, and notifications at each step.

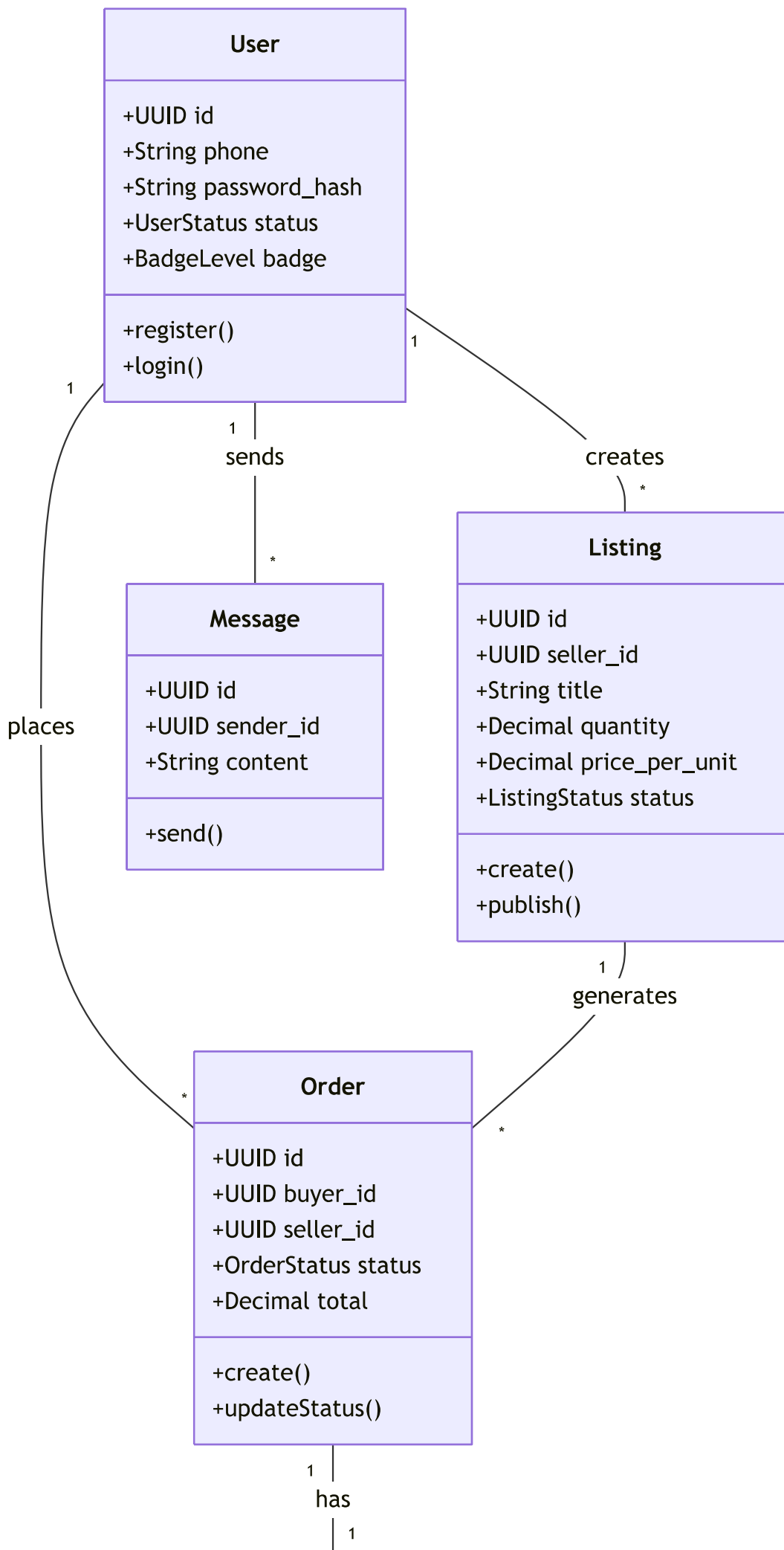
## 4 UML Class Diagram

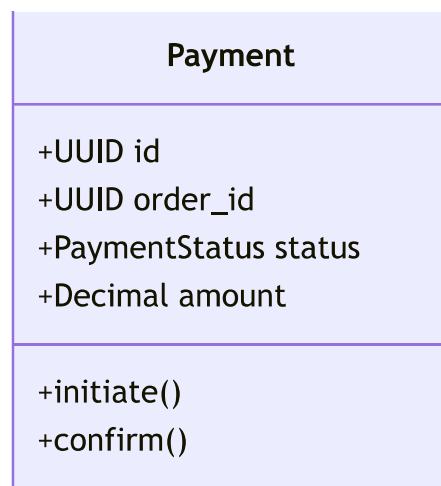
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### Introduction

The class diagram represents the object-oriented structure of the MBOA Market system. It shows the main entities, their attributes, methods, and relationships, following domain-driven design principles.







**Figure 3.4: UML Class Diagram - MBOA Market Domain Model**

### Explanation

The domain model includes: User (authentication and profile), Listing (products for sale), Order (transactions), Payment (mobile money), and Message (chat). Key relationships: A User creates multiple Listings, places multiple Orders, and sends Messages. Each Order has one Payment.

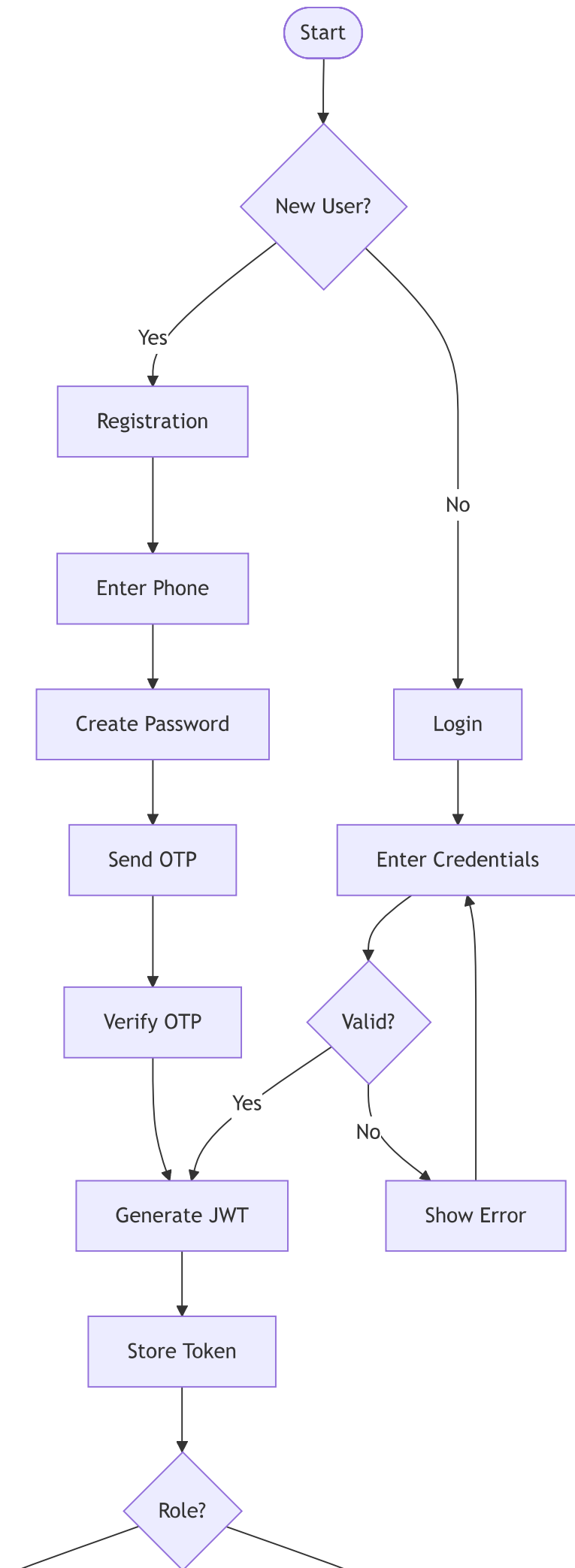
## 5 Flowchart - Authentication Process

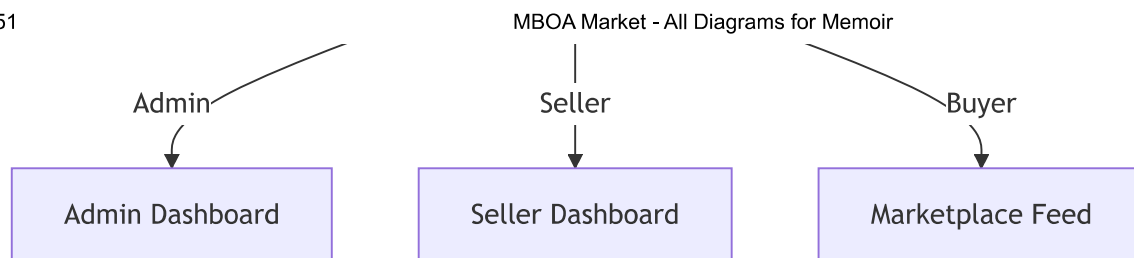
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### Introduction

This flowchart illustrates the complete user journey from initial access to authenticated session. MBOA Market implements a dual-path authentication system handling both new user registration and returning user login.







**Figure 3.5: Flowchart - User Authentication Process**

### Explanation

The authentication flow has two paths: Registration (phone, password, OTP verification) and Login (credentials validation). Post-authentication, users are redirected based on their role: Administrators to Admin Dashboard, Sellers to Seller Dashboard, and Buyers to Marketplace Feed.

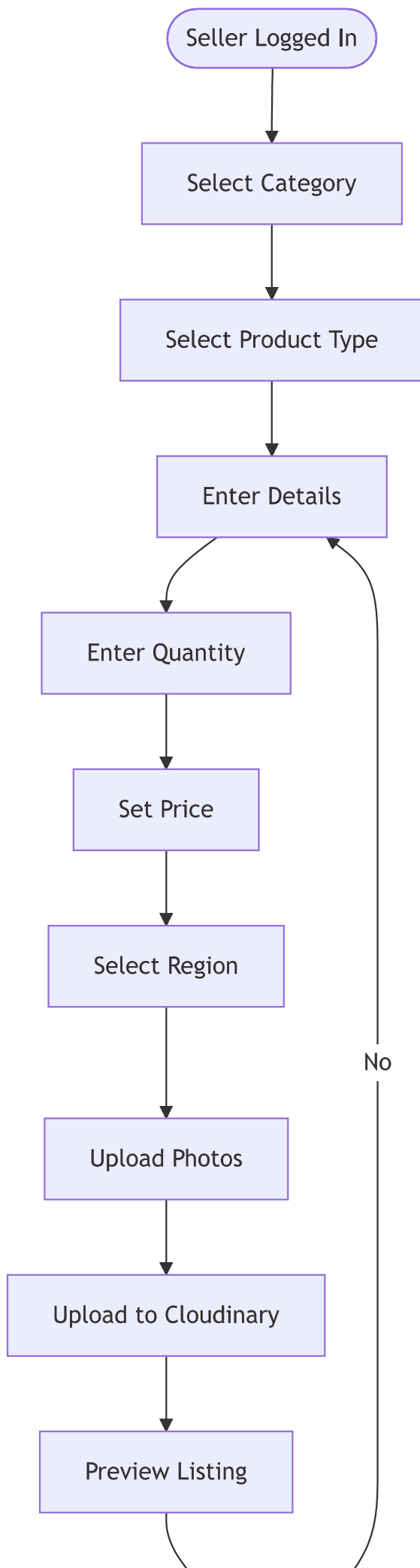
## 6 Flowchart - Listing Creation

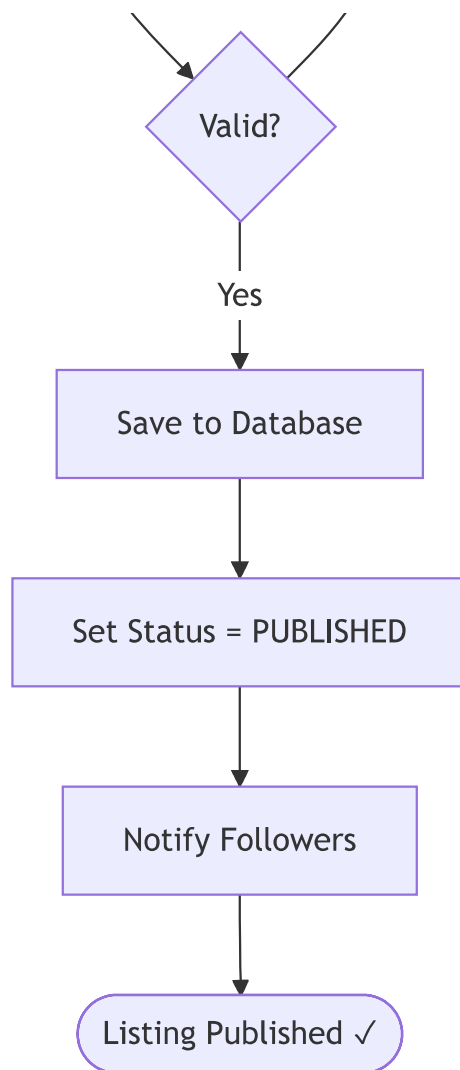
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### Introduction

The listing creation process is the core functionality enabling farmers and breeders to sell their products. This flowchart details the step-by-step process a seller follows to publish a new product listing.







**Figure 3.6: Flowchart - Product Listing Creation Process**

### Explanation

The listing creation has five steps: (1) Product Information (category, type, details), (2) Quantity & Pricing, (3) Location (region, locality), (4) Photos (uploaded to Cloudinary), and (5) Review & Publish. Upon publication, the listing is indexed for search and followers are notified.

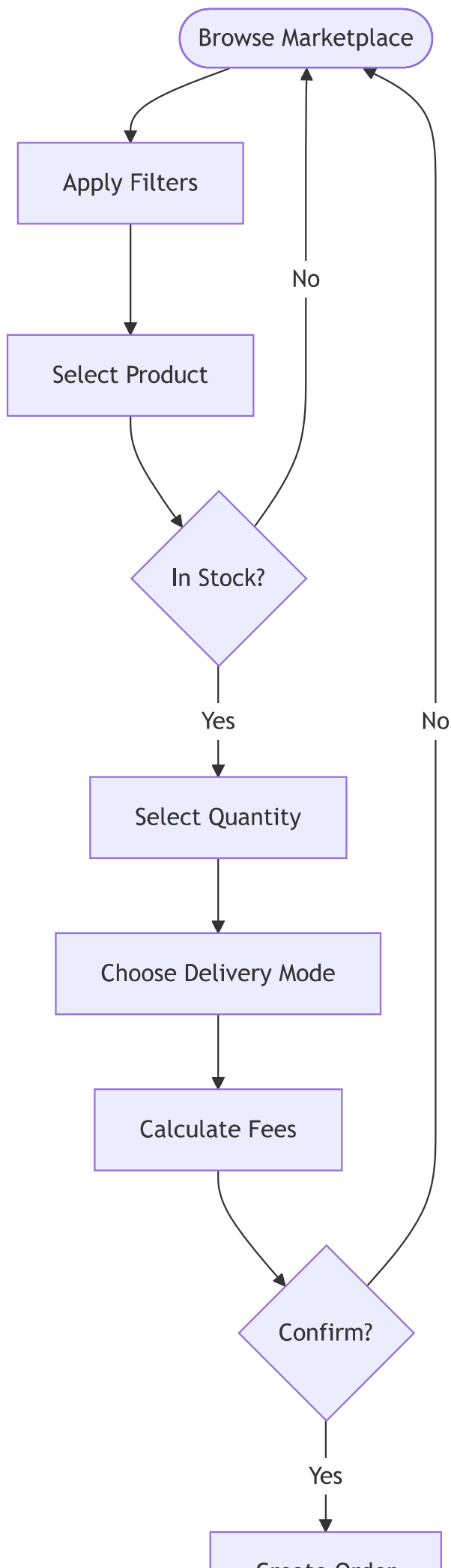
## 7 Flowchart - Order Process

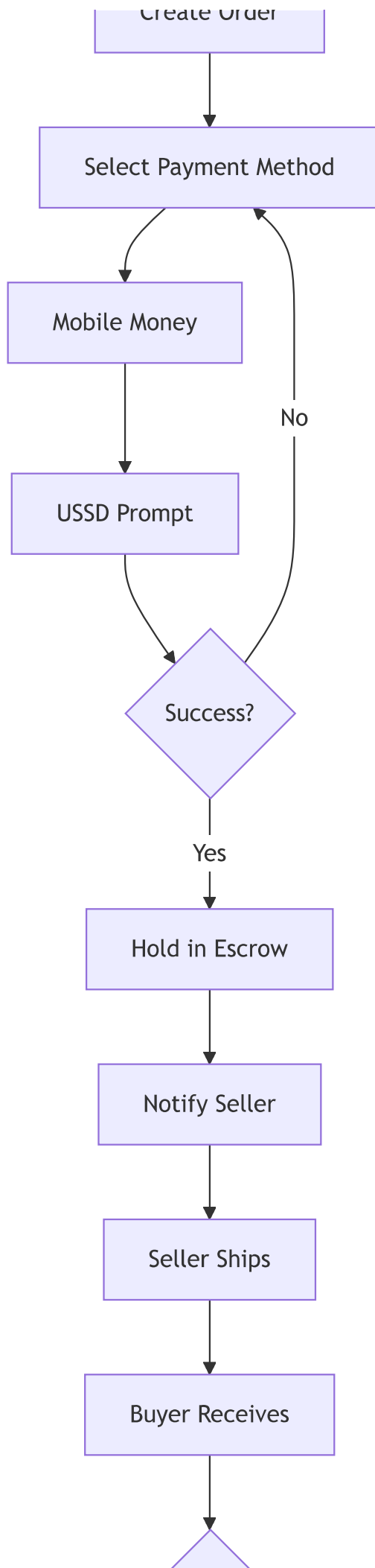
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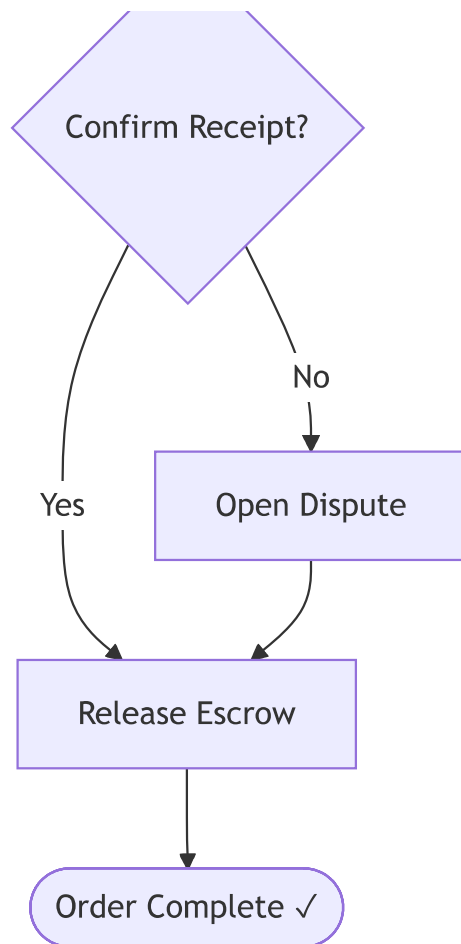
### Introduction

This flowchart illustrates the complete buyer journey from product discovery to order completion. MBOA Market implements an escrow-based payment system protecting both buyers and sellers.









**Figure 3.7: Flowchart - Complete Order Process**

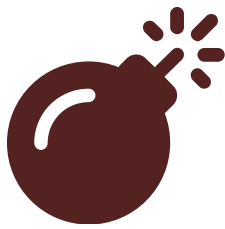
### Explanation

The order process has five phases: (1) Product Discovery with filtering, (2) Optional Negotiation via chat, (3) Order Creation with fee calculation, (4) Payment via Mobile Money with escrow, and (5) Fulfillment with delivery confirmation. Disputes can be opened for admin resolution if needed.

## 8 Component Diagram

### Introduction

The component diagram provides a detailed view of the software components that make up the MBOA Market platform, showing how the system is organized into distinct layers with specific responsibilities.



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*Figure 3.8: Component Diagram - MBOA Market Software Architecture*

### Explanation

Four layers: (1) Client Layer with React, Router, 20 Pages, and API Services, (2) Server Layer with FastAPI, route modules (/auth, /listings, /orders, /ai), and middleware, (3) Data Layer with SQLAlchemy ORM and PostgreSQL, and (4) External Services (Gemini AI, Cloudinary, Mobile Money).

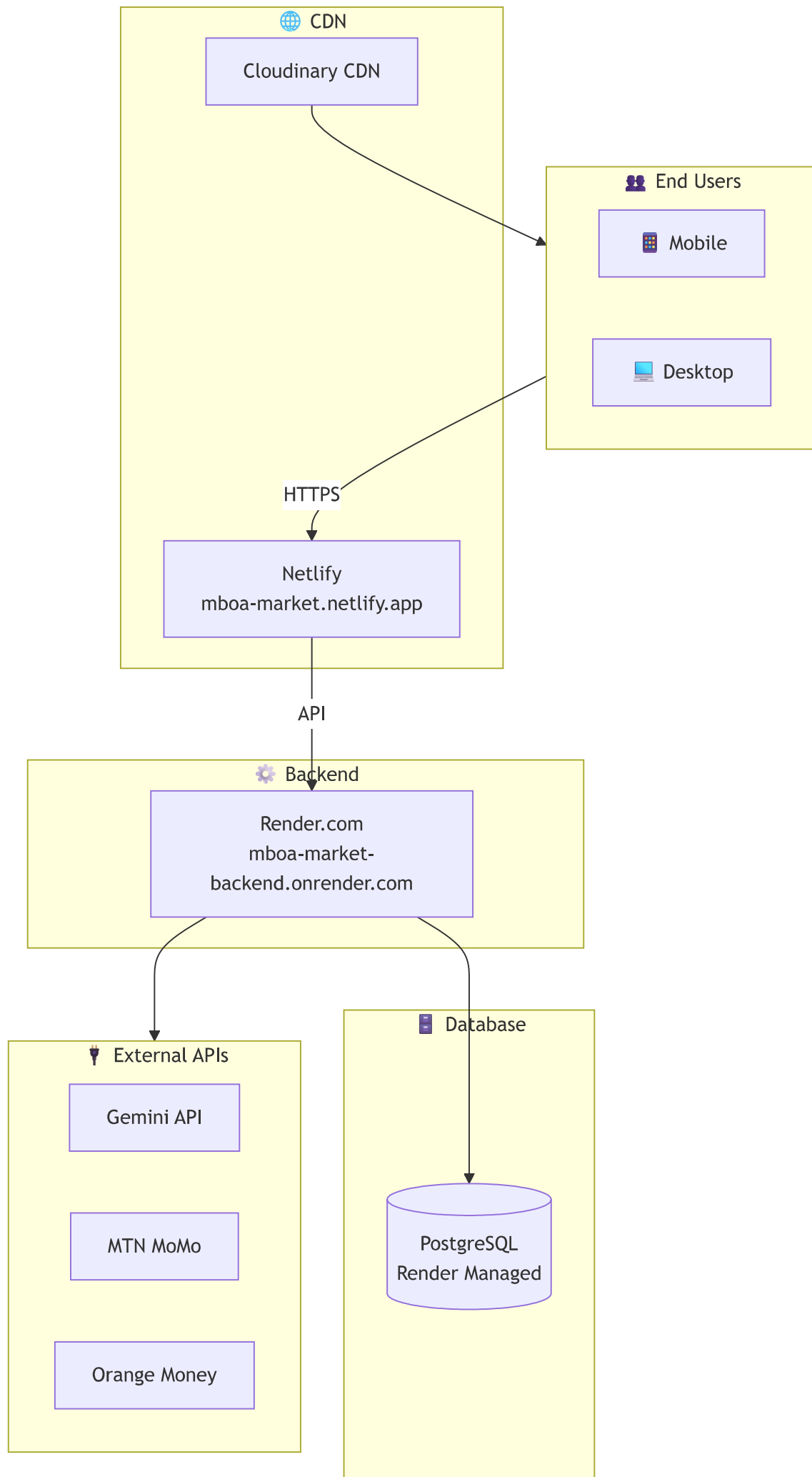
## 9 Deployment Diagram

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### Introduction

The deployment diagram illustrates how MBOA Market is deployed across various cloud services. This architecture leverages modern cloud platforms to ensure scalability, reliability, and global accessibility.





***Figure 3.9: Deployment Diagram - MBOA Market Cloud Infrastructure*****Explanation**

Deployment: Frontend on Netlify (mboa-market.netlify.app), Backend on Render.com (mboa-market-backend.onrender.com), PostgreSQL managed by Render, images via Cloudinary CDN. External APIs: Google Gemini for AI, MTN MoMo and Orange Money for payments. All traffic encrypted via HTTPS.

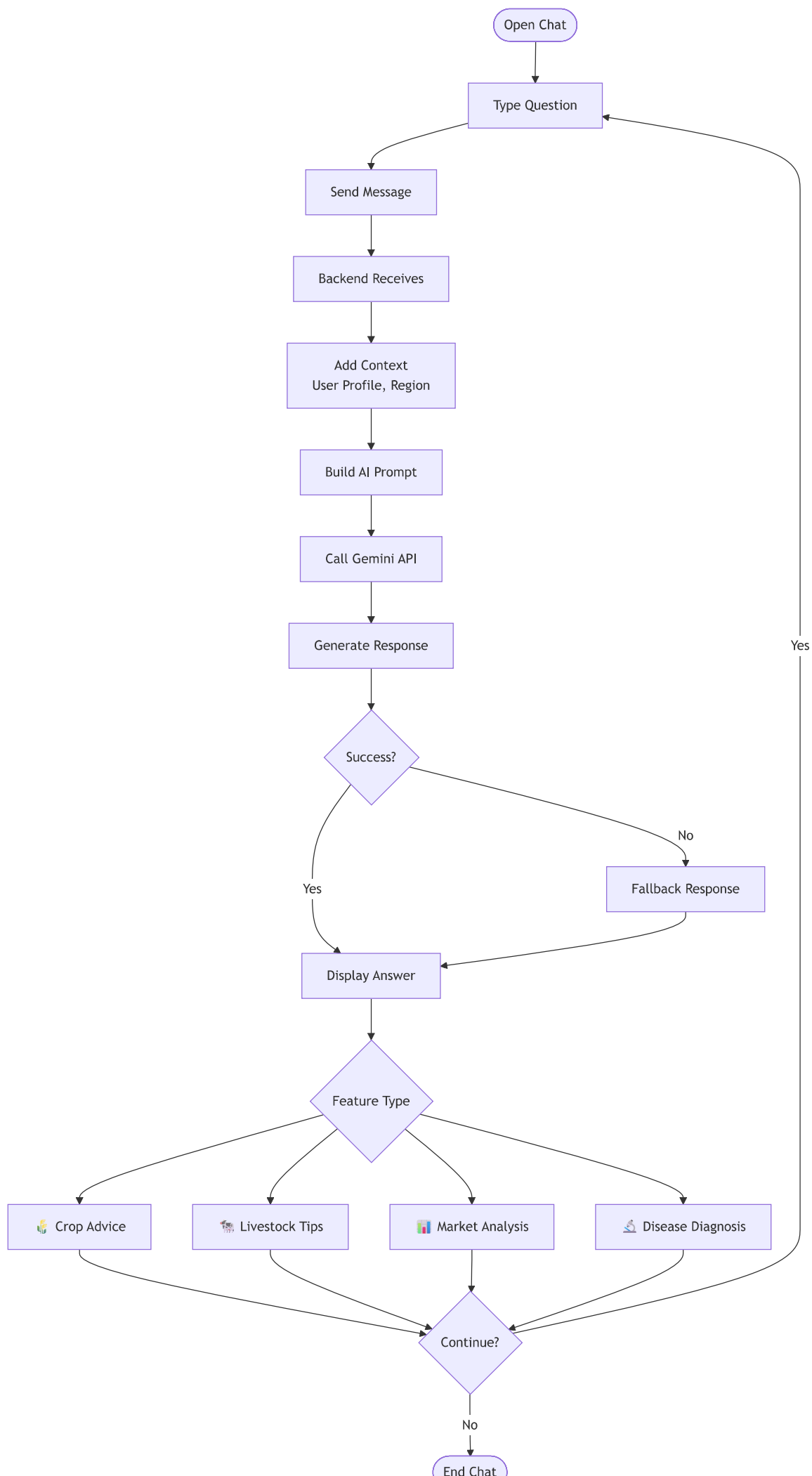
## 10 Flowchart - AI Assistant (Bigiass)

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### Introduction

Bigiass is the AI-powered virtual assistant integrated into MBOA Market, designed to provide agricultural advice and support. This flowchart illustrates how user questions are processed through the Google Gemini AI.





**Figure 3.10: Flowchart - Bigiass AI Assistant Process**

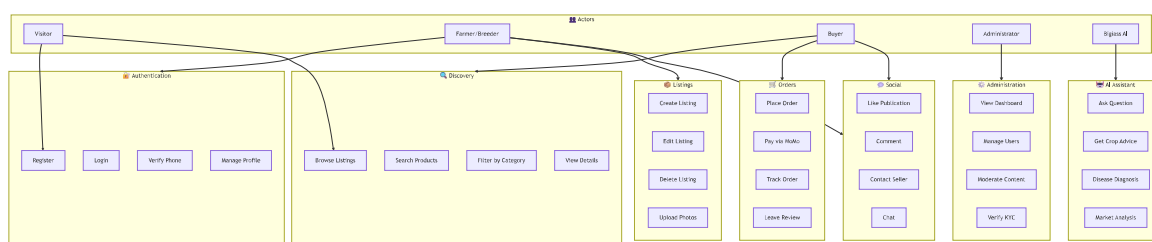
## Explanation

The AI process: (1) User types question, (2) Backend adds context (profile, region), (3) Prompt is built and sent to Gemini API, (4) Response is generated and displayed. Features include: Crop Advice, Livestock Tips, Market Analysis, and Disease Diagnosis. Fallback responses are provided if API fails.

## 11 Use Case Diagram

### Introduction

The use case diagram models the interactions between different actors and the MBOA Market system. It relates five main actors with thirty-seven use cases distributed across seven functional modules.

**Figure 3.11: UML Use Case Diagram - MBOA Market**

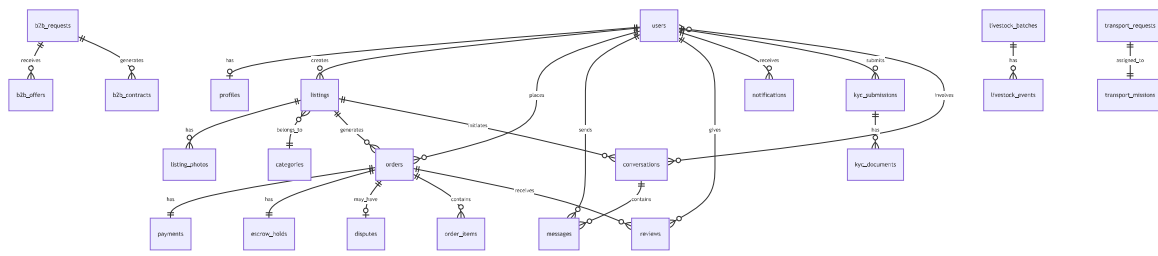
## Explanation

Five actors: Visitor (browse only), Farmer/Breeder (create listings, sell), Buyer (purchase, review), Administrator (manage platform), Bigiass AI (provide advice). Seven modules: Authentication, Listings Management, Discovery, Social Interaction, Orders & Payments, AI Assistant, and Administration.

## 12 Database Entity-Relationship Diagram

### Introduction

The Entity-Relationship Diagram shows the database structure of MBOA Market with 33 tables organized into logical groups. This schema supports all platform functionalities from user management to order processing.



**Figure 3.12: Database Entity-Relationship Diagram - MBOA Market**

### Explanation

The database contains 33 tables organized into groups: User & Auth (users, profiles, roles), Marketplace (listings, categories, photos), Orders & Payments (orders, payments, escrow), Messaging (conversations, messages), KYC (submissions, documents), B2B (requests, offers, contracts), Livestock (batches, events), Logistics (transport requests, missions), and System (notifications, audit logs).